

4.1 Answer each of the following:

- a. Suppose that a simple regression has quantities $N = 20$, $\sum y_i^2 = 7825.94$, $\bar{y} = 19.21$, and $SSR = 375.47$, find R^2 .
- b. Suppose that a simple regression has quantities $R^2 = 0.7911$, $SST = 725.94$, and $N = 20$, find $\hat{\sigma}^2$.
- c. Suppose that a simple regression has quantities $\sum (y_i - \bar{y})^2 = 631.63$ and $\sum \hat{e}_i^2 = 182.85$, find R^2 .

a. $R^2 = \frac{SSR}{SST} = \frac{375.47}{445.458} = 0.8429 *$

$$SST = 7825.94 - 19.21^2 \cdot 20 = 445.458$$

b. $R^2 = \frac{SSR}{SST} = 0.7911 \Rightarrow SSR = 574.2911$

$$SSE = 725.94 - 574.2911 = 151.6489$$

$$MSE = 151.6489 / (20 - 2) = 8.4249 = \hat{\sigma}^2$$

c. $SSR = 631.63 - 182.85 = 448.78$

$$R^2 = 448.78 / 631.63 = 0.7105 *$$

4.3 We have five observations on x and y . They are $x_i = 3, 2, 1, -1, 0$ with corresponding y values $y_i = 4, 2, 3, 1, 0$. The fitted least squares line is $\hat{y}_i = 1.2 + 0.8x_i$, the sum of squared least squares residuals is $\sum_{i=1}^5 \hat{e}_i^2 = 3.6$, $\sum_{i=1}^5 (x_i - \bar{x})^2 = 10$, and $\sum_{i=1}^5 (y_i - \bar{y})^2 = 10$. Carry out this exercise with a hand calculator. Compute

- the predicted value of y for $x_0 = 4$.
- the $se(f)$ corresponding to part (a).
- a 95% prediction interval for y given $x_0 = 4$.
- a 99% prediction interval for y given $x_0 = 4$.
- a 95% prediction interval for y given $x = \bar{x}$. Compare the width of this interval to the one computed in part (c).

a. $\hat{y}_1 = 1.2 + 0.8 \cdot 4 = 4.4$

b. $se(f) = \sqrt{\widehat{Var}(f)} = \sqrt{\hat{\sigma}^2 \left[1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right]} = \sqrt{1.2 \left[1 + \frac{1}{5} + \frac{(4-1)^2}{10} \right]} = \sqrt{2.52} = 1.5875$

$$\hat{\sigma}^2 = MSE = 3.6 \div (5-2) = 1.2$$

$$\bar{x} = (3+2+1-1+0) \div 5 = 1$$

c. 95% P.I. = $\left[\hat{y}_0 \pm t_{0.025}(3) \cdot se(f) \right] = \left[4.4 \pm 3.182 \cdot 1.5875 \right] = [-0.6514, 9.4514]_{**}$

d. 99% P.I. = $\left[\hat{y}_0 \pm t_{0.005}(3) \cdot se(f) \right] = \left[4.4 \pm 5.841 \cdot 1.5875 \right] = [-4.8726, 13.6726]_{**}$

e. 95% P.I. = $\left[2 \pm t_{0.025}(3) \cdot 1.2 \right] = [-1.8184, 5.8184]_{**}$

$$se(f) = \sqrt{1.2 \left[1 + \frac{1}{5} + \frac{(1-1)^2}{10} \right]} = \sqrt{1.44} = 1.2$$

$$\hat{y}_0 = 1.2 + 0.8 \cdot 1 = 2$$

→ The prediction interval is narrowest at the sample mean

$x_0 = \bar{x}$ because the term $(x_0 - \bar{x})^2$ is minimized.

As we move away from the mean, the interval become wider.

4.25 The file *collegetown* contains data on 500 houses sold in Baton Rouge, LA during 2009–2013. Variable descriptions are in the file *collegetown.def*.

- a. Estimate the log-linear model $\ln(PRICE) = \beta_1 + \beta_2 SQFT + e$. Interpret the estimated model parameters. Calculate the slope and elasticity at the sample means, if necessary.
- b. Estimate the log-log model $\ln(PRICE) = \alpha_1 + \alpha_2 \ln(SQFT) + e$. Interpret the estimated parameters. Calculate the slope and elasticity at the sample means, if necessary.
- c. Compare the R^2 value from the linear model $PRICE = \delta_1 + \delta_2 SQFT + e$ to the “generalized” R^2 measure for the models in (b) and (c).
- d. Construct histograms of the least squares residuals from each of the models in (a)–(c) and obtain the Jarque–Bera statistics. Based on your observations, do you consider the distributions of the residuals to be compatible with an assumption of normality?
- e. For each of the models in (a)–(c), plot the least squares residuals against $SQFT$. Do you observe any patterns?
- f. For each model in (a)–(c), predict the value of a house with 2700 square feet.
- g. For each model in (a)–(c), construct a 95% prediction interval for the value of a house with 2700 square feet.
- h. Based on your work in this problem, discuss the choice of functional form. Which functional form would you use? Explain.

5.5 For each of the following two time-series regression models, and assuming MR1-MR6 hold, find $\text{var}(b_2|x)$ and examine whether the least squares estimator is consistent by checking whether $\lim_{T \rightarrow \infty} \text{var}(b_2|x) = 0$.

- a. $y_t = \beta_1 + \beta_2 t + e_t, t = 1, 2, \dots, T$. Note that $\mathbf{x} = (1, 2, \dots, T)$, $\sum_{t=1}^T (t - \bar{t})^2 = \sum_{t=1}^T t^2 - \left(\sum_{t=1}^T t\right)^2 / T$, $\sum_{t=1}^T t = T(T+1)/2$ and $\sum_{t=1}^T t^2 = T(T+1)(2T+1)/6$.
- b. $y_t = \beta_1 + \beta_2 (0.5)^t + e_t, t = 1, 2, \dots, T$. Here, $\mathbf{x} = (0.5, 0.5^2, \dots, 0.5^T)$. Note that the sum of a geometric progression with first term r and common ratio r is

$$S = r + r^2 + r^3 + \dots + r^n = \frac{r(1 - r^n)}{1 - r}$$

- c. Provide an intuitive explanation for these results.

$$\text{Var}(b_2|x) = \text{Var}\left(\frac{\sum (x_t - \bar{x}) y_t}{\sum (x_t - \bar{x})^2}\right) = \frac{\sum (x_t - \bar{x})^2}{\sum (x_t - \bar{x})^4} \text{var}(y_t) = \frac{\sigma^2}{\sum_{t=1}^T (x_t - \bar{x})^2}$$

$$\begin{aligned} \text{a. } x_t &= t, \quad \frac{\sum_{t=1}^T (x_t - \bar{x})^2}{\sum_{t=1}^T (x_t - \bar{x})^2} = \frac{\sum_{t=1}^T (t - \bar{t})^2}{\sum_{t=1}^T (t - \bar{t})^2} = \frac{\sum_{t=1}^T t^2 - \frac{(\sum_{t=1}^T t)^2}{T}}{\sum_{t=1}^T t^2 - \frac{(\sum_{t=1}^T t)^2}{T}} = \frac{T(T+1)(2T+1)/6 - \frac{(T(T+1)/2)^2}{T}}{T(T+1)(2T+1)/6 - \frac{(T(T+1)/2)^2}{T}} \\ &= \frac{T(T+1)(2T+1)}{6} - \frac{T(T+1)^2}{4} = \frac{T^3 - T}{12} \end{aligned}$$

$$\lim_{T \rightarrow \infty} \text{Var}(b_2|x) = \lim_{T \rightarrow \infty} \frac{12\sigma^2}{T^3 - T} = 0 \rightarrow \text{consistent} \quad \#$$

$$\text{b. } \sum_{t=1}^T x_t = 0.5 + 0.5^2 + \dots + 0.5^T = \frac{0.5(1 - 0.5^T)}{1 - 0.5} = 1 - 0.5^T$$

$$\sum_{t=1}^T x_t^2 = 0.5^2 + (0.5^2)^2 + \dots + (0.5^T)^2 = \frac{0.5^2(1 - 0.5^{2T})}{1 - 0.5^2} = \frac{1 - 0.5^{2T}}{3}$$

$$\sum (x_t - \bar{x})^2 = \sum x_t^2 - \frac{(\sum x_t)^2}{T} = \frac{1 - 0.5^{2T}}{3} - \frac{(1 - 0.5^T)^2}{T}$$

$$\lim_{T \rightarrow \infty} \sum (x_t - \bar{x})^2 = \lim_{T \rightarrow \infty} \frac{1 - 0.5^{2T}}{3} - \frac{(1 - 0.5^T)^2}{T} = \frac{1 - 0}{3} - \frac{(1 - 0)^2}{\infty} = \frac{1}{3}$$

$$\Rightarrow \lim_{T \rightarrow \infty} \text{Var}(b_2|x) = \lim_{T \rightarrow \infty} \frac{\sigma^2}{1/3} = 3\sigma^2 \neq 0 \rightarrow \text{not consistent} \quad \#$$

- c. Model (a) is consistent because the regressor $x_t = t$ grows indefinitely, providing "infinite information" as the sample size increases, which drives the estimation error to zero. In contrast, Model (b) is inconsistent because $x_t = 0.5^t$ decays toward zero so rapidly that new data points eventually provide no additional information, causing the total variation S_{xx} to stay finite and the estimation error to never vanish.

5.6 Suppose that, from a sample of 63 observations, the least squares estimates and the corresponding estimated covariance matrix are given by

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \quad \widehat{\text{cov}}(b_1, b_2, b_3) = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

Using a 5% significance level, and an alternative hypothesis that the equality does not hold, test each of the following null hypotheses:

- a. $\beta_2 = 0$
- b. $\beta_1 + 2\beta_2 = 5$
- c. $\beta_1 - \beta_2 + \beta_3 = 4$

a.

$$\textcircled{1} \begin{cases} H_0: \beta_2 = 0 \\ H_a: \beta_2 \neq 0 \end{cases} \quad \textcircled{2} \alpha = 0.05 \quad \textcircled{3} T = \frac{\hat{\beta}_2 - 0}{SE(\hat{\beta}_2)} \sim t(63-3)$$

$$\textcircled{4} RR = \{ |T| \geq t_{0.025}(60) = 2 \} \quad \textcircled{5} T_0 = \frac{3-0}{\sqrt{4}} = 1.5$$

$\textcircled{6} \because T_0 \notin RR$, $\hat{\beta}$ do not reject H_0

$$b. \textcircled{1} \begin{cases} H_0: \beta_1 + 2\beta_2 = 5 \\ H_a: \beta_1 + 2\beta_2 \neq 5 \end{cases} \quad \textcircled{2} \alpha = 0.05 \quad \textcircled{3} T = \frac{\hat{\beta}_1 + 2\hat{\beta}_2 - 5}{SE(\hat{\beta}_1 + 2\hat{\beta}_2)} \sim t(63-3)$$

$$SE(\hat{\beta}_1 + 2\hat{\beta}_2) = \sqrt{3 + 4 + 4(-2)} = \sqrt{11}$$

$$\textcircled{4} RR = \{ |T| \geq t_{0.025}(60) = 2 \} \quad \textcircled{5} T_0 = \frac{2 + 2 \cdot 3 - 5}{\sqrt{11}} = 0.9045$$

$\textcircled{6} \because T_0 \notin RR$, $\hat{\beta}$ do not reject H_0

$$c. \textcircled{1} \begin{cases} H_0: \beta_1 - \beta_2 + \beta_3 = 4 \\ H_a: \beta_1 - \beta_2 + \beta_3 \neq 4 \end{cases} \quad \textcircled{2} \alpha = 0.05 \quad \textcircled{3} T_0 = \frac{\hat{\beta}_1 - \hat{\beta}_2 + \hat{\beta}_3 - 4}{SE(\hat{\beta}_1 - \hat{\beta}_2 + \hat{\beta}_3)} \sim t(63-3)$$

$$SE(\hat{\beta}_1 - \hat{\beta}_2 + \hat{\beta}_3) = \sqrt{3 + 4 + 3 - 2(-2) + 2} = \sqrt{16} = 4$$

$$\textcircled{4} RR = \{ |T| \geq t_{0.025}(60) = 2 \} \quad \textcircled{5} T_0 = \frac{2 - 3 - 1 - 4}{4} = -1.5$$

$\textcircled{6} \because T_0 \notin RR$, $\hat{\beta}$ do not reject H_0

5.20 A generalized version of the estimator for β_2 proposed in Exercise 2.9 by Professor I.M. Mean for the regression model $y_i = \beta_1 + \beta_2 x_i + e_i, i = 1, 2, \dots, N$ is

$$\hat{\beta}_{2,mean} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1}$$

where $(\bar{y}_1, \bar{x}_1)$ and $(\bar{y}_2, \bar{x}_2)$ are the sample means (for the first and second halves) of the sample observations, respectively, after ordering the observations according to increasing values of x . Given that assumptions MR1–MR6 hold:

- Show that $\hat{\beta}_{2,mean}$ is unbiased.
- Derive an expression for $\text{var}(\hat{\beta}_{2,mean} | \mathbf{x})$.
- Write down an expression for $\text{var}(\hat{\beta}_{2,mean})$.
- Under what conditions will $\hat{\beta}_{2,mean}$ be a consistent estimator for β_2 ?

$$a. \quad \hat{\beta}_{2,mean} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1} = \frac{\beta_1 + \beta_2 \bar{x}_2 + \bar{e}_2 - \beta_1 - \beta_2 \bar{x}_1 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1} = \beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}$$

$$\Rightarrow E(\hat{\beta}_{2,mean}) = \beta_2 + \frac{E(\bar{e}_2) - E(\bar{e}_1)}{\bar{x}_2 - \bar{x}_1} \stackrel{=0}{=} \beta_2 \quad \text{[UE]}$$

$$b. \quad \text{Var}(\hat{\beta}_{2,mean} | \mathbf{x}) = \text{Var}\left(\beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}\right) = \text{Var}(\beta_2) + \frac{\text{Var}(\bar{e}_2) + \text{Var}(\bar{e}_1)}{(\bar{x}_2 - \bar{x}_1)^2}$$

$$= \frac{\sigma^2/N/2 + \sigma^2/N/2}{(\bar{x}_2 - \bar{x}_1)^2} = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} \quad \text{✗}$$

$$c. \quad \text{Var}(\hat{\beta}_{2,mean}) = E[\text{Var}(\hat{\beta}_{2,mean} | \mathbf{x})] + \text{Var}[E(\hat{\beta}_{2,mean} | \mathbf{x})]$$

$$= E\left[\frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}\right] \quad \text{✗} \quad \begin{matrix} \beta_2 \\ 4\text{Var}(\beta_2) = 0 \end{matrix}$$

$$d. \quad \lim_{N \rightarrow \infty} \text{Var}(\hat{\beta}_{2,mean} | \mathbf{x}) = \lim_{N \rightarrow \infty} \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} = 0 \rightarrow \text{consistent} \quad \text{✗}$$

$\Rightarrow$ When $N \rightarrow \infty$ ✗

- e. Randomly generate observations on x from a uniform distribution on the interval $(0,10)$ for sample sizes $N = 100, 500, 1000$, and, if your software permits, $N = 5000$. Assuming $\sigma^2 = 1000$, for each sample size, compute:
- $\text{var}(b_2|\mathbf{x})$ and $\text{var}(\hat{\beta}_{2,mean}|\mathbf{x})$ where b_2 is the OLS estimator.
 - Estimates for $E[(s_x^2)^{-1}]$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2]$ where s_x^2 is the sample standard deviation for x using N as a divisor.
- f. Comment on the relative magnitudes of your answers in part (e), (i) and (ii) and how they change as sample size increases. Does it appear that $\hat{\beta}_{2,mean}$ is consistent?
- g. Show that $E[(s_x^2)^{-1}] \xrightarrow{p} 0.12$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2] \xrightarrow{p} 0.16$. [Hint: The variance of a uniform random variable defined on the interval (a, b) is $(b-a)^2/12$.]
- h. Suppose that the observations on x were not ordered according to increasing magnitude but were randomly assigned to any position. Would the estimator $\hat{\beta}_{2,mean}$ be consistent? Why or why not?

$$g. \sigma_x^2 = \frac{(10-0)^2}{12} = \frac{100}{12}$$

$$\textcircled{0} E[(s_x^2)^{-1}] \xrightarrow{p} \frac{1}{\sigma_x^2} = \frac{12}{100} = 0.12$$

$$\textcircled{0} \text{前半部: } x_1, x_2, \dots, x_{N/2} \sim U(0,5) \rightarrow \bar{x}_1 = 2.5$$

$$\text{後半部: } x_{N/2+1}, \dots, x_N \sim U(5,10) \rightarrow \bar{x}_2 = 7.5$$

$$\Rightarrow \bar{x}_2 - \bar{x}_1 = 5$$

$$E\left[\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}\right] \xrightarrow{p} \frac{4}{5^2} = 0.16 \neq$$

- h. If the observations are not ordered, the two subsample means become similar, making the denominator close to zero. This causes the variance to explode, so the estimator is not consistent.

$$e.g. \bar{x}_1 \approx \bar{x}_2, \bar{x}_2 - \bar{x}_1 \rightarrow 0 \Rightarrow \text{Var}(\hat{\beta}_{2,mean}) \rightarrow \infty, \text{ not consistent } \neq$$